

A Multi-Agent Radiology Assistant for Chest X-Ray Abnormality Detection

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Abstract

This project presents a multi-agent pipeline designed as a research tool for detecting abnormalities in chest X-ray images. The system integrates a vision-based binary classifier, a reasoning large language model (LLM), and an automated evaluation agent. The classifier detects abnormal chest X-rays, specifically pneumonia, using a fine-tuned EVA-X Tiny Vision Transformer model. When abnormalities are detected, a reasoning LLM generates radiologic findings and explanations for the prediction. Finally, an evaluation agent assesses the quality of the reasoning output using clinical criteria. The system demonstrates strong recall in pneumonia detection while highlighting challenges in precision and generalization. Overall, the architecture illustrates the potential of combining computer vision models with LLM reasoning in an interactive radiology assistant pipeline.

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1 Problem Description

Detecting abnormalities in medical imaging requires significant expertise. We explore whether a multi-agent AI pipeline can act as a research-oriented assistant for analyzing chest X-rays. The system is designed as a non-clinical tool for exploring how computer vision models and LLMs can collaborate to detect abnormalities and generate interpretable explanations.

The system is designed to:

- classify chest X-ray images as normal or abnormal,

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- generate reasoning-based radiologic explanations, and
- evaluate the quality of those explanations.

2 System Architecture

The platform uses a three-stage multi-agent pipeline:

1. **Classification Agent:** detects whether an X-ray is normal or abnormal;
2. **Reasoning Agent:** generates radiologic findings when abnormalities are detected;
3. **Evaluation Agent:** scores the reasoning output using medical evaluation criteria.

Each agent performs a specialized task while passing context to the next stage in the pipeline. Together, the agents support a Streamlit-based interface that allows users to upload chest X-rays and ask follow-up questions about the results.

3 Agent Design

3.1 Classification Agent

The classification agent uses an EVA-X Tiny Vision Transformer (ViT) [4], with `eva_x_tiny_patch16` pre-trained weights [5] as our backbone, fine-tuned on the Kaggle Chest X-Ray Pneumonia dataset [6]. EVA-X is specifically trained to identify 20 different chest diseases while our model implements a binary classification method (1: Pneumonia, versus 0: normal). We applied preprocessing to make sure the training images were the right size for the Vision Transformer, EVA-X, while also applying data augmentation methods such as `RandomCrop`, `RandomHorizontalFlip` in order to prevent overfitting and improve generalization by diversifying the dataset. The test and validation images were transformed by the `CenterCrop` to reduce inconsistency in the evaluation step. Key model parameters include:

- image size: 224×224 ,
- patch size: 16,
- embedding dimension: 192,
- transformer blocks: 12,
- attention heads: 3.

Each image is divided into 196 patches, which are embedded into 192-dimensional vectors before passing through the transformer layers. Transfer learning was applied by freezing most of the pretrained backbone and fine-tuning only

the final transformer block, normalization layers, and classification head. The classifier outputs the probability p_{abnormal} using a sigmoid activation with binary cross-entropy loss.

Training configuration included:

- optimizer: AdamW,
- learning rate: 1×10^{-4} ,
- weight decay: 1×10^{-4} ,
- early stopping patience: 4 epochs.

The model trained for 13 epochs, with the best performance at epoch 12. In our loss function, we applied a positional weight of 0.7 to address the class imbalance skewing towards a greater number of abnormal training images (1912 Pneumonia vs. 1341 Normal images) to prevent bias towards the majority class. The final output of the binary classifier agent outputs the p_{abnormal} of a chest X-ray image.

3.2 Reasoning LLM Agent

When the classifier predicts abnormality with probability greater than 0.5, a reasoning LLM is invoked to generate radiologic explanations.

The system uses confidence-tiered prompting:

- **0.5–0.7**: emphasizes uncertainty and subtle findings;
- **0.7–0.8**: provides standard findings and a short differential diagnosis;
- **> 0.8**: identifies dominant patterns and likely explanations.

The default reasoning backend is Llama-3.2-11B-Vision-Radiology-mini which was finetuned on the unsloth/Llama-3.2-11B-Vision-Instruct model in order to specialize in “interpreting radiology images such as X-rays, CT scans, and MRIs” [1]. Users may optionally select OpenAI GPT-4.1 [7] instead though it requires their OpenAI key while the local Llama model doesn’t.

This design reduces computational cost and helps avoid hallucinations on clearly normal cases by skipping LLM inference when $p_{\text{abnormal}} \leq 0.5$.

3.3 Evaluation Agent

The third agent evaluates the reasoning output using MedGemma [3] (google/medgemma-2-2b-it).

The evaluation agent scores responses on four criteria:

- correctness,
- relevance,
- safety,
- completeness.

MedGemma is loaded with 4-bit quantization (BitsAndBytesConfig with bnb_4bit_compute_dtype=torch.bfloat16) to fit within GPU memory constraints. The evaluation prompt uses a structured format: the judge receives the context (classifier impression), the question, the LLM’s answer, and a ground truth reference. The response is parsed for JSON containing the scores and justification. In the Streamlit UI,

MedGemma evaluation is an optional toggle in the sidebar (requires GPU and HuggingFace gated model access via HF_TOKEN) Each criterion is scored on a 1–5 scale and returned in structured JSON format.

The evaluation pipeline also includes automated textual metrics such as BLEU and ROUGE, which compare generated responses against reference answers.

The judge returns a structured JSON response with scores and a clinical justification for each evaluation. This provides an automated, reproducible way to benchmark the reasoning LLM’s performance without requiring manual radiologist review.

4 Tools used

The system integrates several open-source models and tools:

- EVA-X Vision Transformer [4],
- Llama-3.2-11B-Vision-Radiology-mini [1],
- MedGemma [3],
- OpenAI GPT-4.1 API [7],
- Streamlit frontend.

Training and experimentation were conducted primarily in Google Colab with GPU acceleration.

5 Experiments

We evaluated the system across three dimensions:

1. classifier training performance,
2. generalization to external images,
3. end-to-end pipeline reasoning quality.

5.1 Experiment 1: Binary Classifier Training and Convergence

Training converged after 13 epochs, with the best model at epoch 12.

Metric	Value
Training Loss	0.0235
Validation Loss	0.0313
Validation Accuracy	100%
Best Epoch	12/13

Table 1. Classifier training results.

5.2 Experiment 2: Classifier Spot Checks on Diverse Images

To assess generalization beyond the Kaggle pneumonia dataset, we tested the trained classifier on external images from Radiopaedia, Physio-pedia, and stock photo sources. Results showed that the model successfully detected abnormal patterns even for out-of-distribution diseases such as tuberculosis and lung cancer.

Image Source	Condition	p_{abnormal}	Prediction
Radiopaedia	Pneumonia	0.9988	Abnormal ✓
Radiopaedia	Normal chest	0.0865	Normal ✓
Physio-pedia	Bronchitis (OOD)	0.9405	Abnormal
Radiology.ca	Tuberculosis (OOD)	0.9986	Abnormal
CU Anschutz	Lung cancer (OOD)	0.9490	Abnormal
NCI	Cancer (OOD)	0.9981	Abnormal
Getty Images	Normal (external)	0.4394	Normal ✓

Table 2. Classifier spot-check results on external chest X-ray images. Checkmarks represent in-distribution classes (Normal, Pneumonia)

The classifier correctly separates clearly normal from clearly abnormal images with high confidence. Out-of-distribution pathologies (bronchitis, TB, lung cancer) are correctly flagged as abnormal, which is expected since the model is trained as a binary normal-vs-abnormal classifier. The Getty normal image ($p=0.44$) falls just below the 0.5 threshold, showing the classifier’s ability to generalize to non-medical-dataset images.

5.3 Experiment 3: Confidence-Tiered LLM Reasoning

When the classifier predicts abnormal, the reasoning LLM generates findings using confidence-tiered prompts. We demonstrate this with the Llama-3.2-11B-Vision-Radiology-mini backend on a known pneumonia case ($p_{\text{abnormal}} = 1.00$, high-confidence tier):

Input: Chest X-ray from Kaggle pneumonia dataset (person1000_bacteria_2931.jpeg) Classifier output: $p_{\text{abnormal}} = 0.9992 \rightarrow$ High confidence tier LLM prompt tier: High — emphasizes dominant patterns and 2–3 likely explanations

LLM output (excerpt): "The provided chest X-ray image [...] The most striking feature in this image is the increased density of the lungs, which is visible as a white or light gray area throughout the thoracic cavity. This indicates that the lungs are filled with a substance that is denser than normal air..."

For normal predictions ($p_{\text{abnormal}} \leq 0.5$), the LLM is not invoked at all, and the system returns only the classification result: "Classifier predicts NORMAL; reasoning model not invoked."

This demonstrates the efficiency design: the LLM (which requires significant compute) is only called when needed.

5.4 Experiment 4: Interactive Q&A Agent

The Streamlit application includes a Q&A chatbot (the "Ask Agent" page) that answers follow-up questions grounded in the inference report context. The Q&A agent receives the full pipeline context — classifier prediction, confidence tier, impression, and LLM reasoning — and uses it to answer user questions about the analysis.

Example interaction flow:

1. User uploads a chest X-ray image
2. Classifier returns: prediction=PNEUMONIA, $p_{\text{abnormal}} = 0.9988$, confidence_tier=high
3. LLM generates radiologic findings describing visible opacities
4. User asks: "What does the confidence tier mean?"
5. Q&A agent responds with an explanation grounded in the specific context, noting the 0.9988 probability and what the high-confidence tier implies

The Q&A agent uses either Llama (local) or OpenAI (gpt-4.1-mini) as the backend, with the same provider selection as the reasoning LLM. The agent’s system prompt explicitly instructs it to be helpful and educational while acknowledging when information isn’t specific to the current image.

6 Results

We evaluated the full pipeline on 60 test images: 30 normal and 30 pneumonia.

6.1 Classification Performance

Metric	Value		
Accuracy	86.7%	Pred. N	Pred. P
Precision	0.789	Normal	22
Recall	1.000	Pneumonia	0
F1 Score	0.882		30

Table 3. Classification performance on the test set (left) and confusion matrix (right).

The classifier achieved perfect recall for pneumonia cases but produced false positives on some normal images.

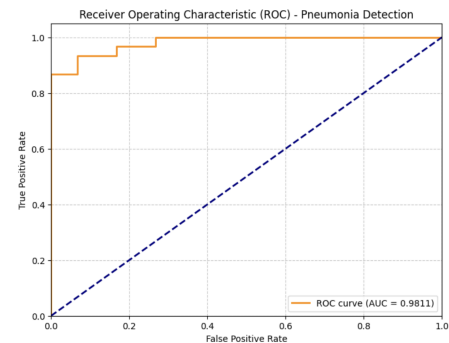


Figure 1. Receiver Operating Characteristic (ROC) curve for pneumonia detection. The classifier achieves an AUC of 0.9811, indicating strong discrimination between normal and pneumonia cases.

Metric	Value
ROUGE-1	0.158
MedGemma Mean Rating	4.02

Table 4. Qualitative Question-Answering Results.

6.2 LLM Evaluation

MedGemma scores correlated with classification correctness, suggesting that the evaluation agent can identify reasoning failures in the pipeline.

6.3 Example Results

Ground Truth	Prediction	p_{abnormal}	Judge Score
NORMAL	PNEUMONIA ×	0.9910	1/5
NORMAL	PNEUMONIA ×	0.9991	1/5
NORMAL	NORMAL ✓	0.2125	5/5
PNEUMONIA	PNEUMONIA ✓	0.9990	4/5
PNEUMONIA	PNEUMONIA ✓	0.9990	4/5
PNEUMONIA	PNEUMONIA ✓	0.9991	4/5

Table 5. Example evaluation results from the radiology assistant pipeline.

6.4 Key observations

- Perfect recall for pneumonia: All pneumonia cases were correctly identified with very high confidence ($p > 0.999$). The judge scored these at 4/5 for correctness, indicating clinically reasonable LLM reasoning
- False positives on normal images: 2 of 3 normal images were incorrectly classified as pneumonia with high confidence ($p > 0.99$). The judge correctly scored these at 1/5, recognizing the mismatch between the LLM’s generated findings and the actual normal image
- Judge alignment: The MedGemma judge scores correlate well with classification correctness — correct predictions received scores of 4–5/5, while false positives received 1/5. Validates the evaluation agent’s ability to catch errors in the reasoning pipeline
- Precision-recall tradeoff: The 100% recall / 78.9% precision pattern suggests the classifier is biased toward predicting abnormality. Expected given the class imbalance in training data (1,912 pneumonia vs. 1,341 normal)

6.5 Limitations of These Results

- The Kaggle test set’s normal images may have characteristics that differ from the training distribution, contributing to false positives
- The evaluation used Gemma-2-2B-it as the judge rather than the larger MedGemma-1.5-4B-it model due to

Colab memory constraints, which may affect scoring quality

- ROGUE quantitative scores were extremely poor when comparing MedGemma answers to questions to the OpenAI model’s answers. This is somewhat expected due to the non-deterministic nature of LLMs generally

7 Discussion and Limitations

Several limitations affect the system’s performance:

- limited dataset diversity beyond pneumonia,
- class imbalance in the training data,
- hardware constraints limiting larger-model evaluation,
- non-deterministic behavior in LLM-generated answers.

Future improvements include:

- expanding the training data to include multiple thoracic diseases,
- testing larger and more specialized medical foundation models,
- scaling experiments with additional compute resources,
- improving precision while preserving high recall.

8 Conclusion

This project demonstrates the feasibility of integrating computer vision classification, LLM reasoning, and automated evaluation into a multi-agent radiology assistant pipeline. Although the system is not suitable for clinical use, it highlights how AI agents can collaborate to analyze medical imaging data and generate more interpretable reasoning about model predictions.

GitHub Repository

https://github.com/rithika-d/Big_Data_Analytics_Midterm_Project [2]

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